

Amazon Detective

Amazon Detective is one of the coolest new features in AWS that helps in remediating security issues across AWS workloads by using a combination of statistical analysis and machine learning analysis. Detective collects logs from various AWS resources and runs its intelligence to create interactive visualisations and reports, which can be used by customers to detect security anomalies with ease. It enhances the use of CloudWatch by combining log data analysis with alerts when it gets access to sensitive resources like APIs, URLs, etc.

Amazon CodeGuru

AWS has introduced a machine learning based automated code review service that can help in reviewing code changes and also find performance bottlenecks like concurrency issues, duplicate codes, resource leaks, input validations, and many more.

AWS CodeGuru can be an event watcher for the Git repository or the AWS CodeCommit repository. It scans the code intelligently for standards violations by doing a pull request, and shares a report of the findings with solutions to fix the issues. AWS CodeStar is another service which helps developers handle workspaces for code development. CodeGuru can be an integral part of this toolchain activity to automate code review, and build and deploy an application through a pipeline.

AWS CodeGuru has two integral components – ‘Reviewer’ to enable code review and provide solutions, and ‘Profiler’ to analyse the runtime behaviour of applications and provide recommendations for fixing performance bottlenecks.

Code Reviewer typically helps to associate with any code repository like AWS CodeCommit or Git, and carries out static code analysis on a selected branch to find a potential resource leak or code quality issues, as shown in Figure 2.

Profiler can run Java code in AWS to find potential quality issues including

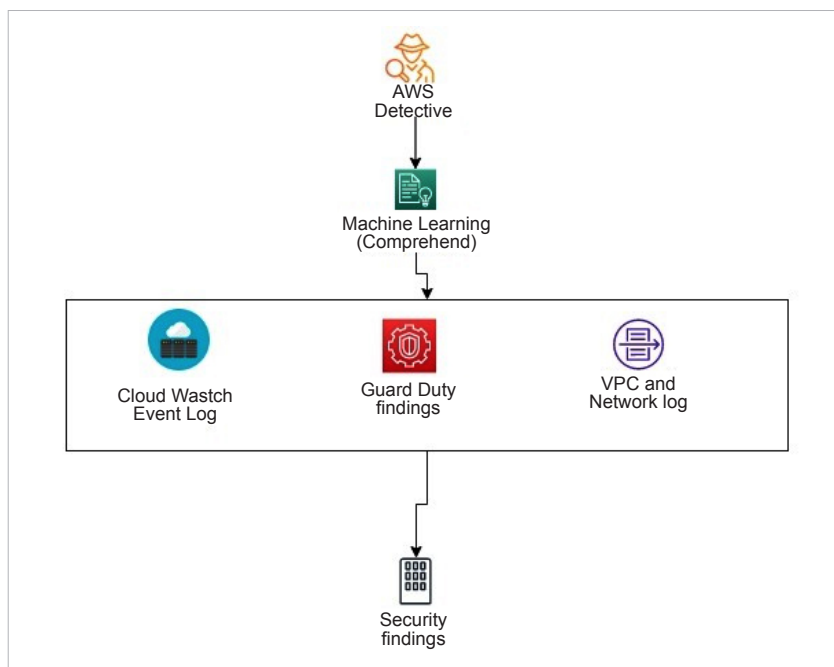


Figure 1: Amazon Detective

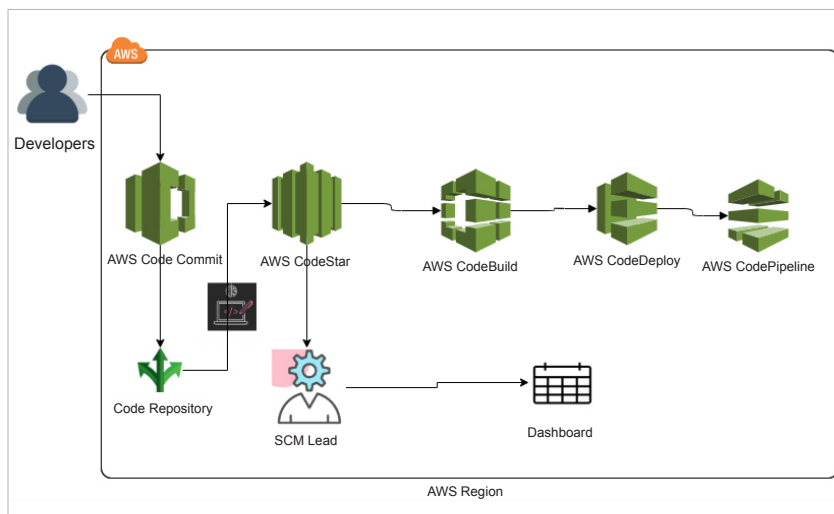


Figure 2: Amazon CodeGuru

resource leaks, memory inefficiency, code optimisation and reusability. Reviewer and Profiler can be associated with a CI/CD pipeline in order to automate static code analysis.

Amazon Serverless Bot Framework

Implementing a chatbot or conversational bot is important in many fields like customer support

and helplines. AWS offers prebuilt CloudFormation templates from Marketplace to swiftly develop a serverless chatbot service.

A typical architecture of this kind of conversational bot using the AWS Lambda function and Polly service is shown in Figure 3. Lambda is a serverless function to implement the business logic. S3 hosts the Web application (user interface for chatbot interface), Cognito is for